

Lab Practice: Decision Trees and Ensemble Trees

Learning Objectives

- Understand the fundamental concepts of decision tree learning.
- Develop the ability to extract decision rules from tree-based models.
- Implement decision tree and ensemble learning using Python.

Exercise 1

You are evaluating job candidates based on the following characteristics:

- Technical Expertise $\in \{\text{Low, Medium, High}\}$,
- Communication Skills $\in \{\text{Weak, Strong}\}$,
- Degree $\in \{\text{BSC, MSC, PhD}\}$.

Our training dataset to determine if a candidate is suitable (S) or not suitable (NS) for a job might look like this:

Technical Expertise	Communication Skills	Degree	Candidate Suitability
High	Strong	BSc	S
Medium	Strong	BSc	NS
High	Weak	PhD	NS
Low	Weak	BSc	NS
Medium	Strong	MSc	S
High	Strong	MSc	S
High	Strong	PhD	S
Medium	Weak	BSc	NS
Low	Weak	MSc	NS
High	Strong	PhD	S

Information Gain = Entropy(parent) – [average Entropy(children)]

Entropy = $\sum_i -p_i \log_2 p_i$, p_i is the probability of the class i

1. Construct a decision tree using the ID3 algorithm. Select the attributes based on the highest information gain at each step.
2. Derive two decision rules from the constructed decision tree, clearly indicating the conditions leading to a specific classification.

Entropy of parent:

Total sample = 10
positive = 5
negative = 5

$$-\left(\frac{5}{10} \log_2 \frac{5}{10} + \frac{5}{10} \log_2 \frac{5}{10}\right) = 1.0 \rightarrow \text{prob of being a parent class.}$$

Technical Skill

(5) high: $-\frac{4}{5} \log_2 \frac{4}{5} - \frac{1}{5} \log_2 \frac{1}{5} = 0.7219$

(3) medium: $-\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} = 0.9183$

(2) low: $-\frac{0}{2} \log_2 \frac{0}{2} - \frac{2}{2} \log_2 \frac{2}{2} = 0$

$$\frac{5}{10} \times 0.7219 + \frac{3}{10} \times 0.9183 + \frac{2}{10} \times 0 = 0.6364$$

(2)

$$1 - 0.6364 = 0.3636$$

Common Skill

(6) strong: $-\frac{5}{6} \log_2 \frac{5}{6} - \frac{1}{6} \log_2 \frac{1}{6} = 0.65$

(4) weak: $-\frac{0}{6} \log_2 \frac{0}{6} - 1 \log_2 1 = 0$

$$1 - \left(\frac{6}{10} \times 0.65\right) = 0.41$$

(1)

Degree

(4) BSc: $-\frac{1}{4} \log_2 \frac{1}{4} - \frac{3}{4} \log_2 \frac{3}{4} = 0.8113$

(2) MSc: $1 \log_2 1 + 0 \log_2 0 = 0$

(4) PhD: $-\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} = 1$

$$-\frac{2}{3} \log_2 \left(\frac{2}{3}\right) - \frac{1}{3} \log_2 \left(\frac{1}{3}\right)$$

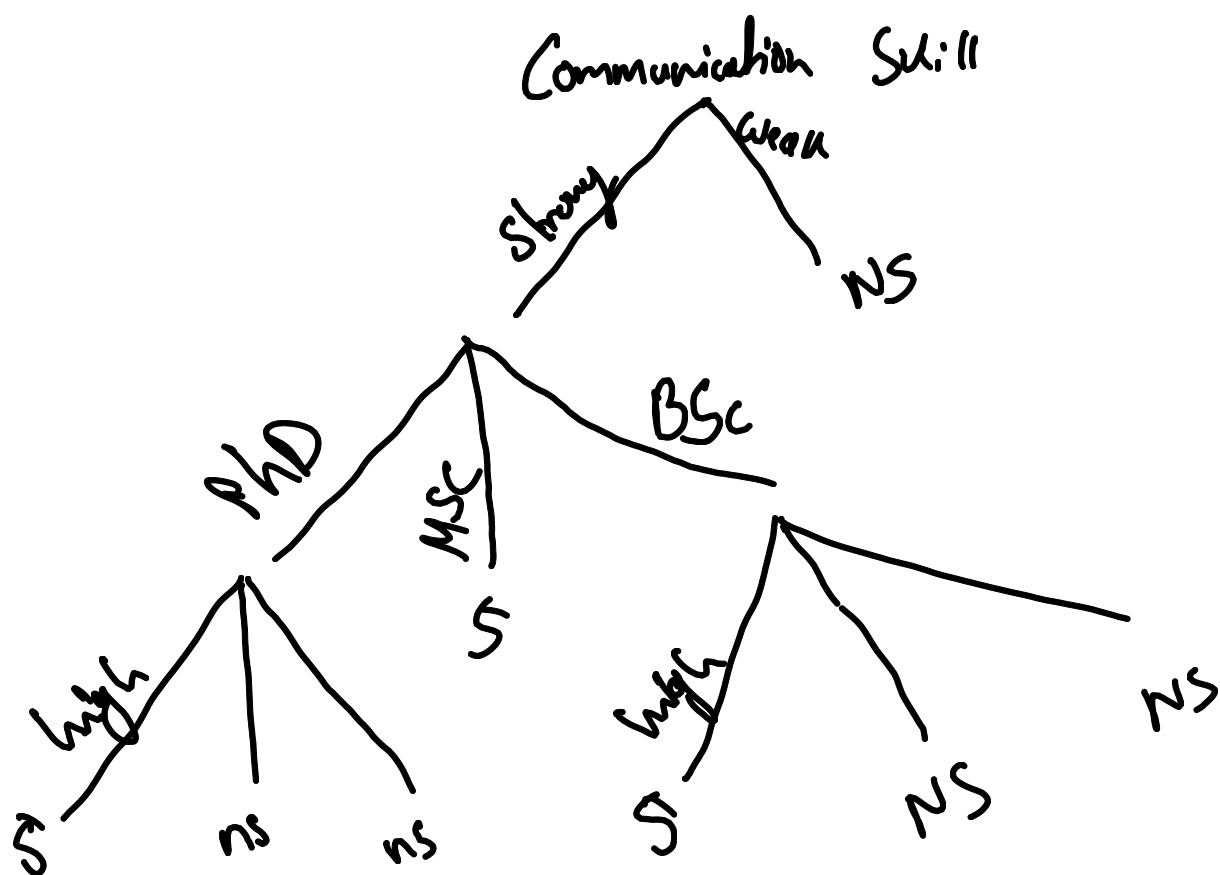
$$-\frac{2}{3} \log_2 \left(\frac{2}{3}\right) - \frac{1}{3} \log_2 \left(\frac{1}{3}\right)$$

$$\frac{4}{10} \times 0.8113 + 0 + 1 \times \frac{4}{10} = 0.7285$$

$$1 - 0.7285 = 0.2715$$

(3)

$$1 - 0.676 = 0.324$$



if comms = weak NS

if comms = strong $\exists$ degree MSc = S

Exercise 2

Step 1: Install Required Libraries

Install pandas, numpy, matplotlib, seaborn, and scikit-learn using pip.

Step 2: Load the Dataset

Use pandas to read the heart.csv dataset and explore the structure and content of the data.

Step 3: Handle Missing Values & Outliers

Check for null values and remove outliers using Z-score to ensure clean data.

Step 4: Encode Categorical Features

Convert categorical columns (like Sex, ChestPainType, etc.) to numerical values using LabelEncoder.

Step 5: Feature Scaling

Normalize the feature values using StandardScaler for better model performance.

Step 6: Split the Dataset

Split the data into training and testing sets using train_test_split.

Step 7: Train a Decision Tree Classifier

Fit a basic DecisionTreeClassifier.

Step 8: Apply Bagging with Random Forest Classifier

Use bagging with a random forest classifier.

Step 9: Apply Boosting with XGBoost

Apply Boosting with XGBoost.

Step 10: Compare Performance

Compare the accuracy of the basic decision tree, bagging, and boosting versions to analyze the improvement.